

FARAZ CHISHTY

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EDUCATION:

Rutgers University | New Brunswick, NJ

May 2025 Bachelor of Science in Mechanical Engineering

- **Coursework:** Dynamic Systems and Control, Design of Mechanical Components, Thermodynamics, Heat Transfer

EXPERIENCE:

Research Assistant | Rutgers University | New Brunswick, NJ

Feb. 2024 - May 2024

- **Scrubbing Robot:** Assisted in developing a new cleaning robot by researching tendon-driven systems for effective cleaning of soft and uneven surfaces.
- **Design Documentation:** Analyzed PID control strategies, evaluated pneumatic and hydraulic systems, and documented implementation details to improve positional accuracy and reduce project costs of the scrubbing robot.

PROJECTS:

Dice Rolling Robot

Jan. 2026 - Feb. 2026

- **Digital Dice-Roller:** Built an interactive robot that simulates dice rolls via **touchscreen input**, using randomized outputs, servo motion, and audio feedback to serve as an engaging alternative to traditional dice.
- **Modular Design:** Fabricated **3D printed components** using **OnShape** to streamline assembly and refinement.
- **Root Cause Analysis:** Diagnosed erratic behavior during testing by isolating subsystems and identifying insufficient current from the power source; solved the issue by integrating lithium-ion batteries to meet servo current demands.

Surfboard Propulsion Mechanism - Capstone

Sept. 2024 - May 2025

- **Mechanical Waverider:** Designed and engineered a surfboard-mounted propulsion system using **SolidWorks**, machining, and assembly to assist beginner surfers in catching waves.
- **Material Selection:** Conducted research on **lightweight, waterproof materials** to ensure durability, corrosion resistance, and manufacturability under variable load and saltwater conditions.
- **Structural Analysis:** Conducted **stress analysis** to determine maximum bending stress on the system's paddle, evaluating alternative materials (HDPE and neoprene) to maximize force to dollar ratio.
- **Force Generation:** Integrated and modified a high-efficiency winch system, **achieving an output to input force ratio of about 9:1** force pounds and making it easier for surfers to catch waves.
- **Project Documentation:** Documented theory and assembly processes in a comprehensive report and presented the results of our testing through live presentations.

Autonomous Targeting System

Jun. 2024 - Aug. 2024

- **Game-Triggered Turret:** Created a face-tracking turret that fires toy projectiles at a user when they lose a videogame.
- **Controls Design:** Programmed a control system in **Python** and **Arduino** for automated targeting with 95% accuracy, demonstrating cross-disciplinary integration of mechanical, electrical, and software components.
- **Facial Recognition Algorithm:** Implemented **facial tracking** algorithms with webcam input to position a servo-controlled toy gun toward a target and fire upon text cues via **Optical Character Recognition**.
- **Iterative Design:** Refined the initial prototype by designing and integrating custom 3D-printed components.

ORGANIZATIONS:

President | RU BREAKS

May 2022 - May 2025

- **Event Planning & Management:** Organized the largest breakdancing competition on the East Coast, securing a Red Bull sponsorship, and coordinating logistics for famous judges and dancers nationwide.
- **Outreach:** Increased breakdancing club membership from 6 to 40+ by creating a national competitive team and a performance team to showcase talent at campus events, boosting exposure and attracting new members.

SKILLS:

- CAD (SolidWorks, OnShape), FEA (ANSYS), Python, MATLAB, Simulink, Arduino IDE, Microsoft Office, OpenCV, Arduino, Raspberry Pi, Soldering, Circuits, Material Selection, Stress & Structural Analysis, Laser Cutting, Woodshop, 3D Printing (Bambu Lab P1S)